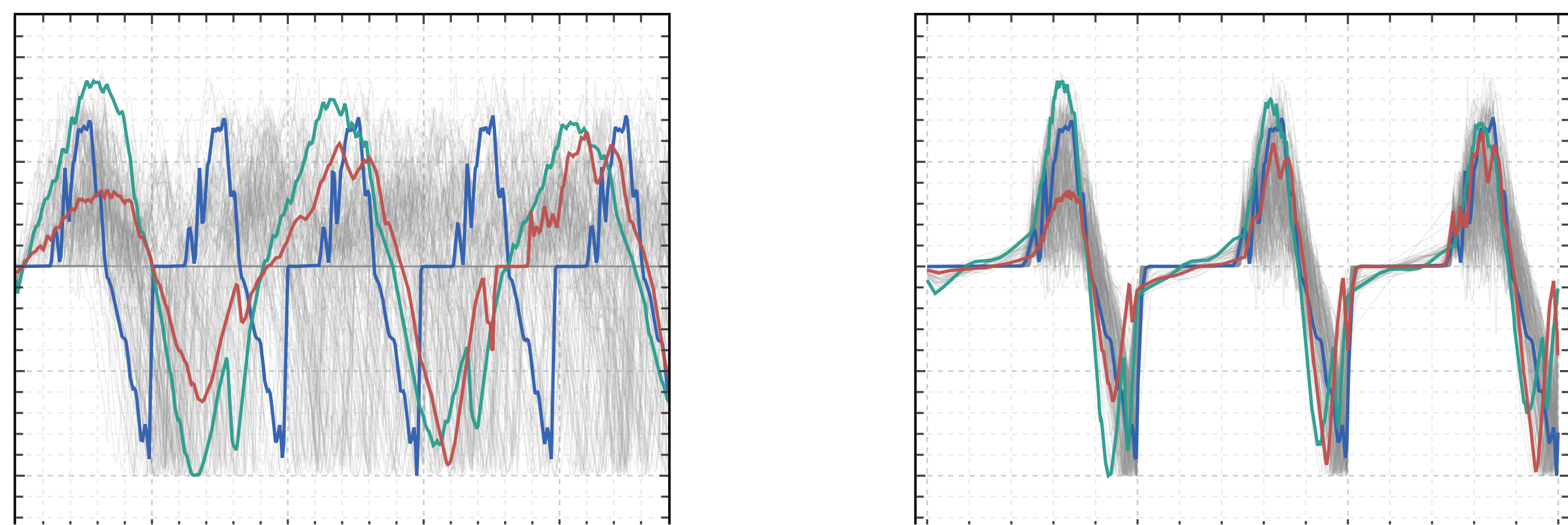


WANTED: SIMULATIONS IN NEED OF A SURROGATE

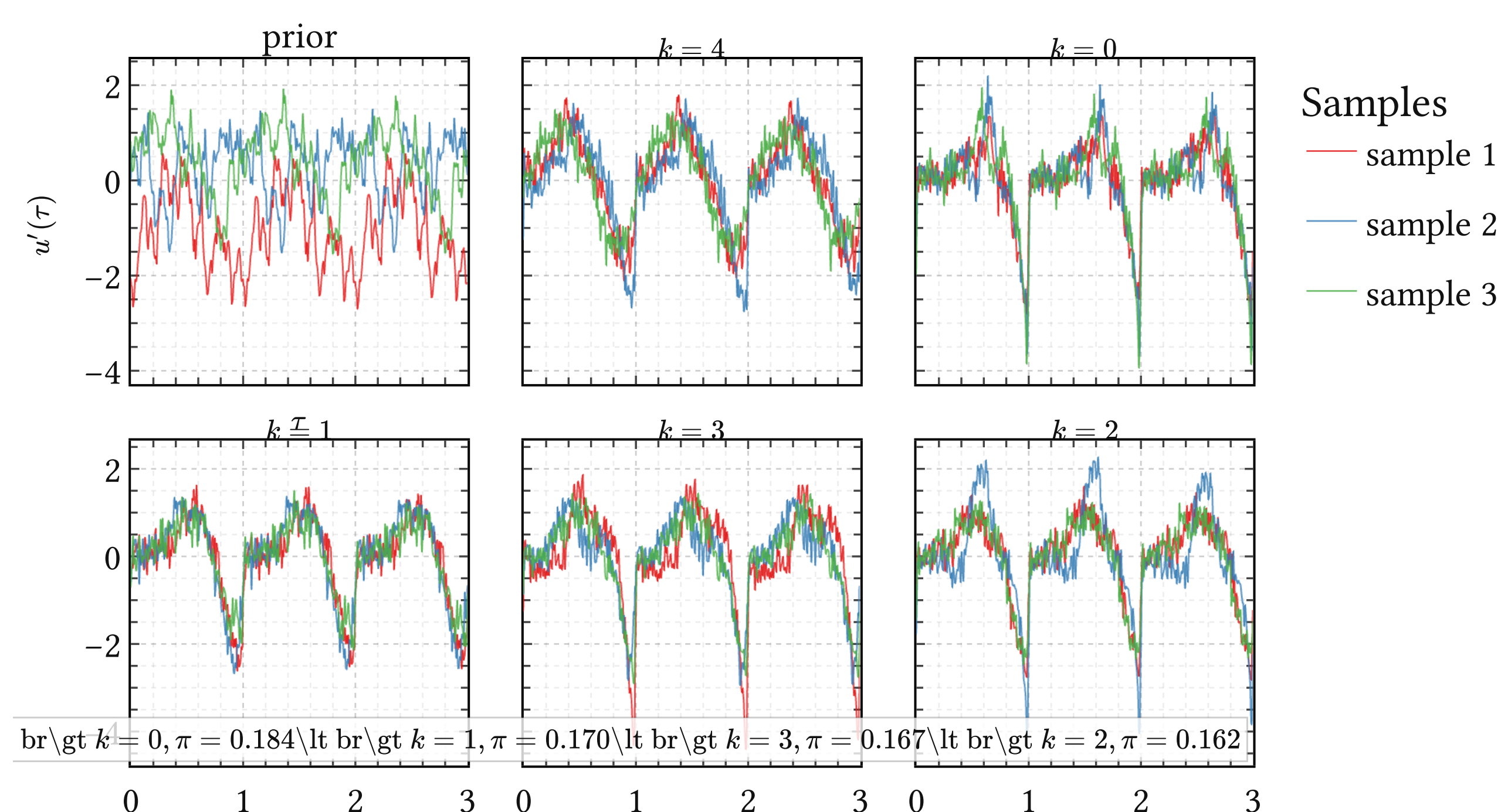
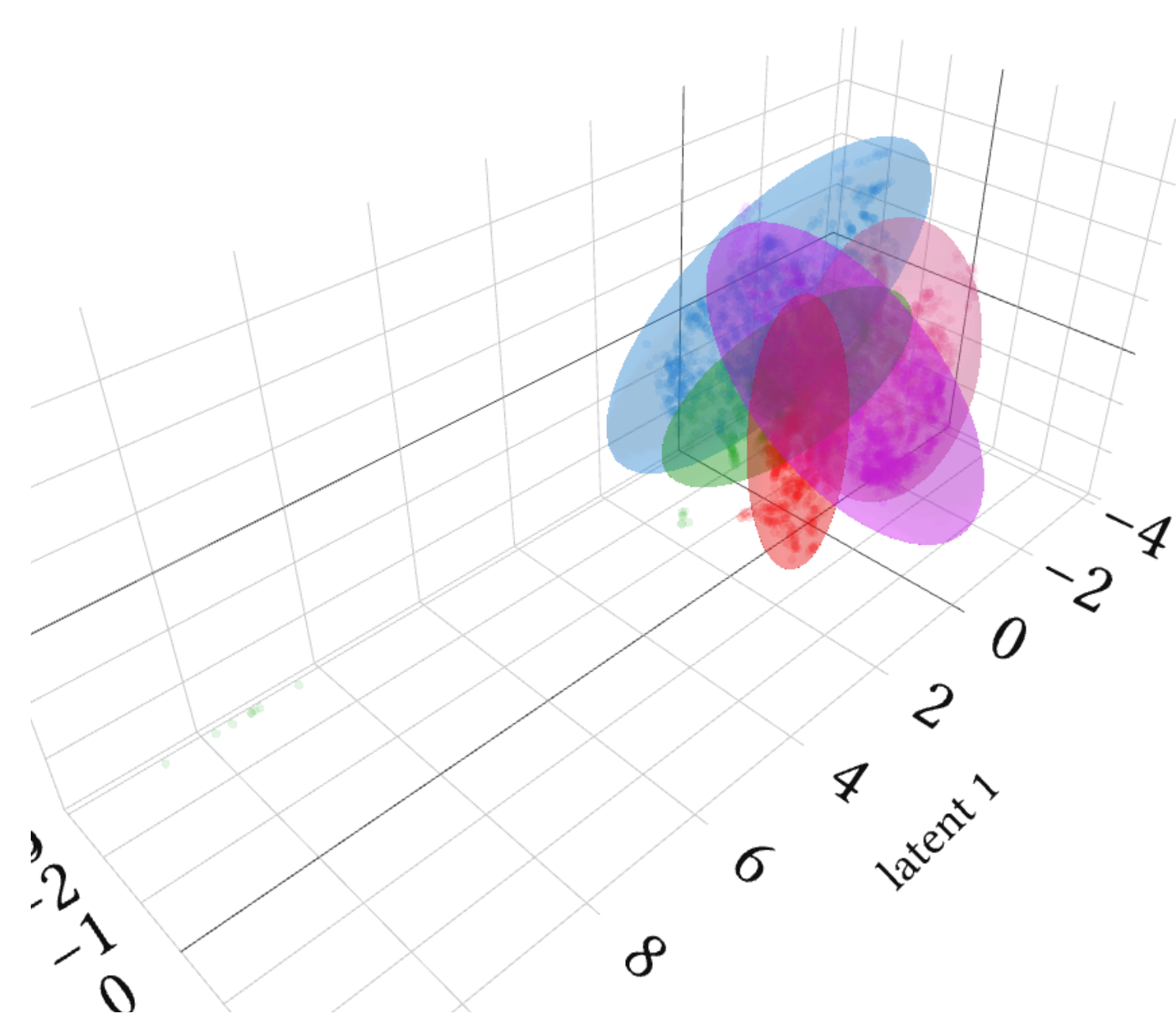
THOUSANDS OF SLOW SIMULATIONS...

High-fidelity vocal fold simulations: thousands of airflow waveforms, hours of compute, one cycle each after registration.



...BECOME THREE NUMBERS AND ERROR BARS

A Bayesian GPLVM on a learned sparse basis: every waveform gets a point on a latent manifold with its own uncertainty, and a handful of archetypes explains the corpus.



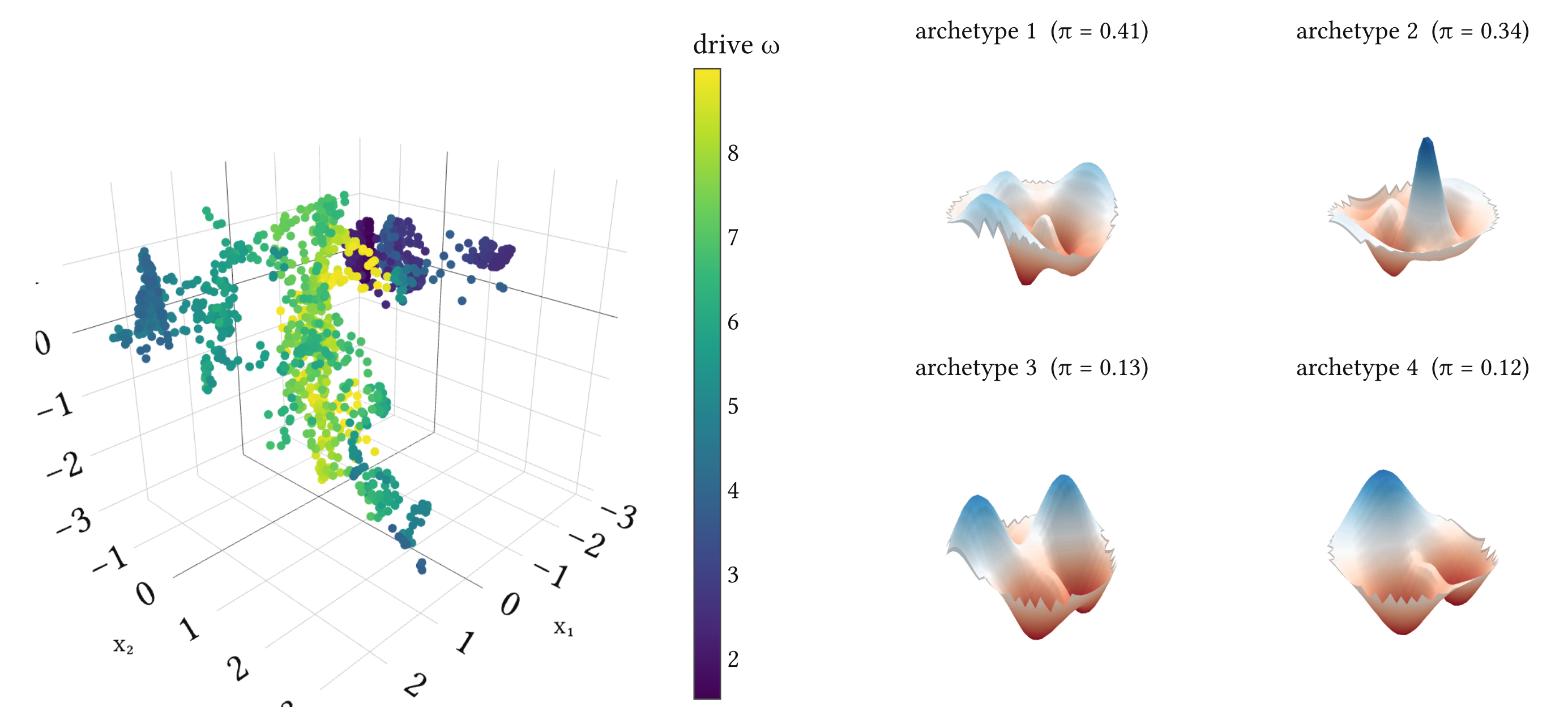
IT IS NOT ABOUT SPEECH

Side quest for this poster: 1,600 point-driven membrane simulations (drive frequency, force position, damping). Same pipeline, untouched.

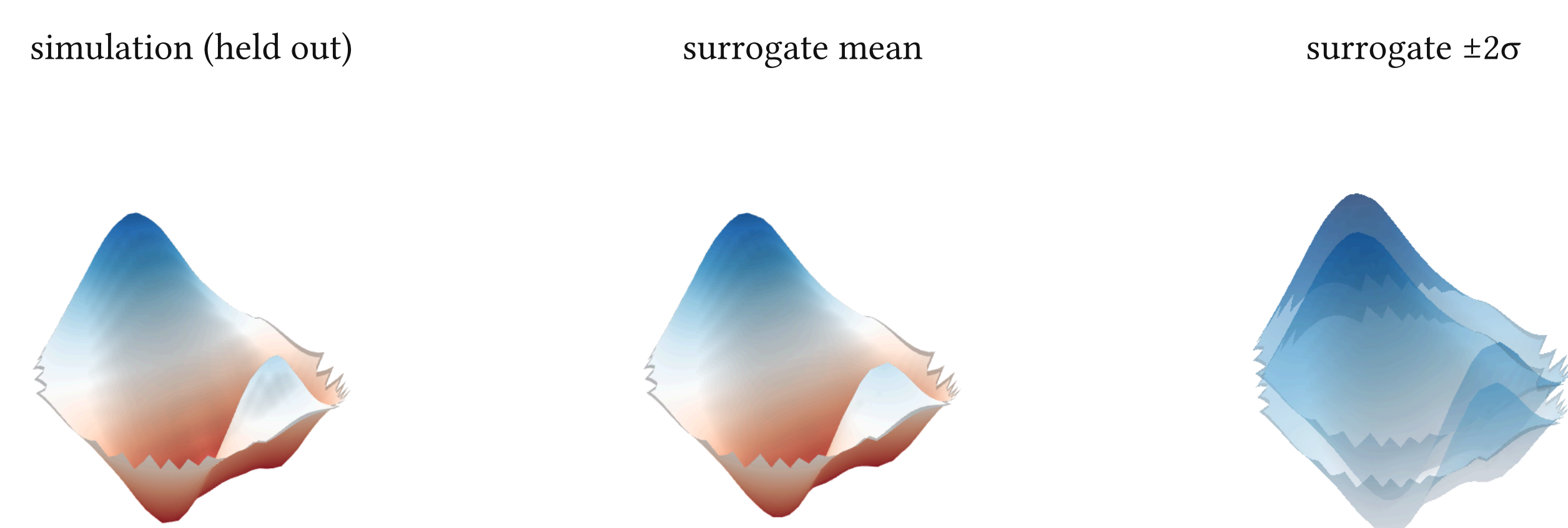
$\omega = 2.4$ $\omega = 4.7$ $\omega = 1.9$ $\omega = 4.5$



Four of the simulations.



Latent manifold coloured by drive frequency, and the four archetypes the surrogate learned.



A held-out simulation (a median case), its reconstruction from three numbers (8% error), and the $\pm 2\sigma$ band.

YOUR SIMULATOR HERE

Finite-element models, vibrometry, parameter sweeps you cannot afford to redo. Bring them: variability becomes a probability distribution you can compute with.

